

Nonlocal Unification Field Theory and Its Quantum Optics Limit: Observer-Function Formulation

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We formulate the *Nonlocal Unification Field Theory* (NUFT) by treating the observer function $f(\Delta I)$ as the primary field generating spacetime geometry from informational differentials. This aligns with the Nonlocal Construct (NC) formalism in which geometry is emergent from smooth, bounded observer mappings rather than pre-existing manifolds. The resulting variational framework yields coupled field equations that unify gravitational and informational dynamics. In the weak-curvature limit, NUFT reduces naturally to quantum optics, where the curvature of the observer function generates optical squeezing and nonlinearity. This observer-function formulation smooths the conceptual transition between cosmological NU dynamics and laboratory-scale optical phenomena.

I. INTRODUCTION

The *Nonlocal Unification* (NU) framework [1] posits that spacetime is not fundamental but constructed by observer-dependent mappings between informational differentials ΔI and measurable intervals $(\Delta x, \Delta t)$:

$$(\Delta x, \Delta t) = f(\Delta I). \quad (1)$$

Here $f : \mathbb{R} \rightarrow \mathbb{R}^{1,3}$ is a smooth, monotonic, embodiment-dependent *observer function* that resides in a Fréchet-topologized function space $O \subset C^r(M, \mathbb{R}^{1,3})$, as established in [2]. The geometry perceived by an observer arises as the Cauchy limit f_∞ of a filtration $\{f_n\}$, corresponding to increasing informational resolution.

Our aim here is to construct a covariant field theory whose dynamical variable is $f(\Delta I)$, and to demonstrate that in the small-curvature limit it reproduces quantum-optical field dynamics.

II. OBSERVER FUNCTION AS PRIMARY FIELD

We define the emergent metric as a functional of the observer function:

$$g_{\mu\nu}(f) = \partial_\mu f(\Delta I) \partial_\nu f(\Delta I), \quad (2)$$

which encodes the rate of informational sampling. Smoothness and boundedness of f guarantee the absence of curvature singularities (No Singularity Theorem).

Let ∇_μ denote the covariant derivative compatible with $g_{\mu\nu}(f)$. We introduce the *informational curvature tensor*

$$\mathcal{F}_{\mu\nu} = \nabla_\mu f_\nu - \nabla_\nu f_\mu, \quad f_\mu \equiv \nabla_\mu f. \quad (3)$$

This tensor measures the nonlocal twist of observer sampling across information gradients.

III. ACTION AND LAGRANGIAN DENSITY

We propose the observer-function Lagrangian

$$\mathcal{L}_{\text{NU}} = \frac{c^3}{16\pi G}(R - 2\Lambda) - \frac{\lambda}{4}\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu} - V(f, \partial f, \Delta I), \quad (4)$$

with action

$$S[f] = \int d^4x \sqrt{-g(f)} \mathcal{L}_{\text{NU}}. \quad (5)$$

The potential $V(f, \partial f, \Delta I)$ captures nonlinear dependence on the observer function and its gradient with respect to ΔI . Variation of S with respect to f and $g_{\mu\nu}$ yields the NU field equations.

A. Field Equations

Variation with respect to the metric gives

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}^{(f)}, \quad (6)$$

$$T_{\mu\nu}^{(f)} = \lambda(\mathcal{F}_{\mu\alpha}\mathcal{F}_\nu{}^\alpha - \frac{1}{4}g_{\mu\nu}\mathcal{F}_{\alpha\beta}\mathcal{F}^{\alpha\beta}) - g_{\mu\nu}V + \frac{\delta V}{\delta g^{\mu\nu}}. \quad (7)$$

Variation with respect to f yields

$$\nabla_\mu(\lambda \mathcal{F}^{\mu\nu}) + \frac{\partial V}{\partial f_\nu} - \nabla_\mu \left(\frac{\partial V}{\partial(\nabla_\mu f_\nu)} \right) = 0. \quad (8)$$

Equations (6)–(8) form the **Informational Einstein–Maxwell system** expressed directly in terms of $f(\Delta I)$.

IV. REDUCTION TO QUANTUM OPTICS

To recover laboratory-scale physics, we linearize around a flat metric:

$$g_{\mu\nu}(f) \rightarrow \eta_{\mu\nu}, \quad f(\Delta I) = \alpha\Delta I + \beta(\Delta I)^3 + \dots \quad (9)$$

The linear term reproduces the standard Minkowski metric, while cubic corrections generate optical nonlinearities.

A. Linear (Coherent) Limit

Neglecting β and higher-order terms,

$$\partial_\mu \partial^\mu f + \frac{\partial V}{\partial f} = 0. \quad (10)$$

For $V(f) = \frac{1}{2}\omega_0^2 f^2$, this reduces to the Klein–Gordon equation. Upon quantization, excitations of f correspond to *informational photons* or *infons*, identical in algebra to optical field quanta.

B. Nonlinear (Squeezed) Regime

Including $\beta(\Delta I)^3$ yields an interaction Hamiltonian

$$H_{\text{int}} \propto \beta(a^{\dagger 2} + a^2), \quad (11)$$

producing the two-photon squeezing term familiar in quantum optics. Thus, squeezing and Kerr nonlinearities arise from curvature in the observer function $f(\Delta I)$.

V. INTERPRETATION AND CONSISTENCY

a. Relation to NU Foundations. The present field theory is consistent with the appendices of the *Consequences of Nonlocal Unification for Limit Observers* paper. In that formalism, f is the generator of the metric, and the entropy differential ΔI is the conserved informational substrate. Boundedness of $\partial^k f$ ensures the absence of singularities, directly embodying the No Singularity Theorem. The action above corresponds to the variational principle $S[f] = \int L(f, \partial f, \Delta I) d^4x$ defined over the Fréchet space O .

b. Physical correspondence. The linear optical limit corresponds to small deviations of f from its mean observer configuration, while cosmological behavior corresponds to large-scale evolution of f across embeddings in O . The same governing equations, interpreted at different scales, reproduce both smooth cosmic expansion and coherent optical propagation.

c. Experimental sector. Quantum-optical experiments probe perturbations of $f(\Delta I)$ at the laboratory

scale, manifesting as field amplitude fluctuations. Gravitational or cosmological observations correspond to large-scale variations of f constrained by bounded entropy ΔI . The shared origin of both sectors under NUFT offers a unified epistemic field theory.

VI. CONCLUSION

The Nonlocal Unification Field Theory in observer-function form establishes a continuous bridge from informational geometry to quantum optics. By elevating the observer function $f(\Delta I)$ to primary field status, the formalism unifies curvature, information flow, and optical field dynamics within a single variational framework. The weak-field limit reproduces quantum optics; the strong-field regime yields cosmological regularity and the No Singularity Theorem. This duality positions NUFT as both a conceptual and practical unification of informational and physical theories.

VII. ACKNOWLEDGMENT

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Appendix A: Mathematical Supplement: Fréchet Structure

Let $O \subset C^r(M, \mathbb{R}^{1,3})$, $r \geq 2$, denote the space of observer functions. Equip O with seminorms $\|f\|_{k,K} = \sup_{x \in K} |D^k f(x)|$ for compact $K \subset M$. (O, τ_O) is complete and locally convex; every Cauchy filtration $\{f_n\}$ converges to $f_\infty \in O$ generating smooth, singularity-free metrics $g_{\mu\nu}(f_\infty)$. This construction grounds the field theory in the same mathematical domain described in the NU appendices.

Appendix B: Informational Regularity and Energy

Define the informational energy density $E = \frac{1}{2}\|\nabla f\|^2 + V(f, \partial f, \Delta I)$. Under bounded derivatives, $E(T) \leq E(0)e^{CT}$, ensuring finite curvature and energy throughout informational time T .

Appendix C: Quantum Optics Correspondence

Linearizing f and identifying $E \leftrightarrow$ electric field quadrature yields the standard quantized Hamiltonian of optical modes. Curvature-induced deviations of f map to squeezing parameters, connecting laboratory quantum optics directly with NU observer geometry.

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